

Rational Matrix Policy Optimization for Rational Feed-Forward Language Models

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Abstract

Rational feed-forward blocks introduce useful nonlinear structure, but they also change the geometry seen by an optimizer. We study Rational Local Basis (RLB) feed-forward layers in Transformer language models and introduce Rational Matrix Policy Optimization, an optimizer that treats the input and output matrices around each rational block as role-specific objects rather than generic dense weights. The method combines a standard base optimizer for the rest of the model with role-depth scaling, on-policy group statistics, and a bounded gauge rebalance for the RLB matrices. A preliminary three-seed real-corpus screen on FineWeb and FineWeb-Edu shows about a 0.15 validation-loss improvement over same-protocol SiLU+AdamW and the strongest non-MatrixPolicy control in that screen. These results motivate a full optimizer benchmark with tuned baselines, speed-to-target curves, stability accounting, and scale/transfer tests.

1 Introduction

Most language-model optimizer studies treat the feed-forward block as a generic collection of matrices. That is appropriate for standard elementwise activations, but it misses structure when the feed-forward layer is itself a learned rational map. In an RLB block, one matrix chooses the normalized domains presented to rational groups, the rational coefficients shape each group response, and a second matrix recombines the resulting features into the residual stream. These roles create optimizer-visible structure that is not captured by a single scalar learning-rate rule.

This paper studies the following claim: rational feed-forward networks expose geometry that a block-aware optimizer can exploit, and an optimizer that uses that geometry can train rational language models more robustly than generic optimizers under the same base protocol. The method is Rational Matrix Policy Optimization (MatrixPolicy). It keeps the surrounding Transformer training recipe unchanged and modifies only the optimization of matrices adjacent to RLB groups. The central mechanism is not that rational activations are intrinsically better than SiLU; it is that the rational parameterization gives the optimizer additional structure, including a positive scale gauge and group-level pressure signals.

This work makes three contributions. First, it defines the RLB matrix roles and the MatrixPolicy update used in our language-model experiments. Second, it reports a preliminary but replicated real-corpus screen on FineWeb and FineWeb-Edu, including divergent plain-RLB controls rather than hiding them. Third, Appendix B gives a narrow theoretical motivation: RLB blocks have a scale gauge, coordinatewise adaptive updates are not equivariant to this gauge, and MatrixPolicy’s rebalance step acts as a bounded canonicalization of the representative. The appendix is deliberately not a convergence theorem; the empirical benchmark remains the basis for claims about validation loss.

2 Background and Related Work

Adam-style adaptive methods normalize coordinatewise gradient moments and remain the default optimizer family for many Transformer language-model runs (Kingma & Ba, 2015).

Recent optimizer papers have raised the evidence standard for new optimizers: fair tuning over baseline families, throughput and memory accounting, speed-to-target curves, stability reporting, and transfer beyond the selection setting are now expected. MatrixPolicy should be judged against that standard, not against a single default learning rate.

Rational feed-forward layers are a different axis from generic optimizer design. They replace a fixed elementwise nonlinearity with learned rational maps applied to normalized groups. This gives the optimizer access to quantities that do not exist in a SiLU FFN: group radii, rational-coefficient pressures, input/output matrix roles, and a positive gauge relating the adjacent matrices. MatrixPolicy is therefore a structure-specific optimizer. It is complementary to broad optimizer families such as AdamW, Muon-style matrix updates, schedule-free methods, factored adaptive methods, and matrix-preconditioned methods; the final paper benchmark must tune and compare against those families directly.

3 Rational Matrix Policy Optimization

Let $\mathcal{L}(\theta)$ be the language-model training objective. For layer l , the RLB block projects the residual-stream input $x_l \in \mathbb{R}^d$ into G groups of width m ,

$$z_l = A_l x_l, \quad z_{l,g} = \text{group}_g(z_l), \quad (1)$$

normalizes each group by its RMS radius,

$$r_{l,g} = \sqrt{m^{-1} \|z_{l,g}\|_2^2 + \epsilon_{\text{rms}}}, \quad u_{l,g} = z_{l,g} / r_{l,g}, \quad (2)$$

and applies a learned rational map in normalized coordinates,

$$h_{l,g} = r_{l,g} R_{l,g}(u_{l,g}), \quad y_l = B_l \text{concat}_{g=1}^G h_{l,g}. \quad (3)$$

The input matrix A_l selects rational domains, the coefficients of $R_{l,g}$ shape the group response, and the output matrix B_l recombines rational features. MatrixPolicy uses this separation by partitioning parameters as

$$\theta = \theta_{\text{base}} \cup \theta_{\text{coeff}} \cup \{A_l, B_l\}_{l=1}^L. \quad (4)$$

The base parameters use the same optimizer as the non-rational baseline. Rational coefficients use a coefficient rule. MatrixPolicy is applied only to A_l and B_l .

In the homogeneous case $\epsilon_{\text{rms}} = 0$, each group has a positive scale gauge:

$$A_{l,g} \mapsto a_g A_{l,g}, \quad B_{l,g} \mapsto a_g^{-1} B_{l,g}, \quad a_g > 0. \quad (5)$$

The represented function is unchanged because $z_{l,g}$ and $r_{l,g}$ scale together, $u_{l,g}$ is unchanged, and the inverse scale in B_l cancels the scale in $h_{l,g}$. With a nonzero RMS floor, this symmetry is approximate but remains locally controlled under mild nondegeneracy assumptions (Appendix B). This matters because optimizer state is stored in coordinates, not in the quotient space of represented functions.

At step t , each matrix role $r \in \{\text{in}, \text{out}\}$ receives a role-depth factor $\rho_r(l)$ and an on-policy statistic factor $s_{\text{stat}}(l, r, t)$. The matrix update is

$$\begin{aligned} M_{l,r}^{t+1} = & M_{l,r}^t + \Delta_{\text{adapt}}(M_{l,r}; \eta_t a_{\text{mat}}(l, r, t) [1 - \mu(l, r, t)]) \\ & + \Delta_{\text{mat}}(M_{l,r}; \eta_t a_{\text{matnorm}} \mu(l, r, t)), \end{aligned} \quad (6)$$

where $M_{l,\text{in}} = A_l$, $M_{l,\text{out}} = B_l$, η_t is the shared base learning rate, Δ_{adapt} is an Adam-style adaptive step, and Δ_{mat} is a short early matrix-normalized step. The mixture coefficient $\mu(l, r, t)$ is active only early in training. After the matrix update, MatrixPolicy computes a target log-ratio for each group and applies the bounded rebalance

$$A_{l,g} \leftarrow e^{\ell_g} A_{l,g}, \quad B_{l,g} \leftarrow e^{-\ell_g} B_{l,g}. \quad (7)$$

The update changes the representative used by the optimizer while preserving, or with $\epsilon_{\text{rms}} > 0$ approximately preserving, the represented block function. Full hyperparameter definitions are in Appendix A.

Table 1: Preliminary three-seed real-corpus screen. Gaps are validation-loss improvements for RLB+MatrixPolicy over the comparison row.

Task	MP loss	Std.	PPL	Gap vs. SiLU	Gap vs. ctrl
FineWeb	4.369701	0.026358	79.04	0.159263	0.152302
FineWeb-Edu	4.069422	0.002281	58.52	0.154149	0.153402

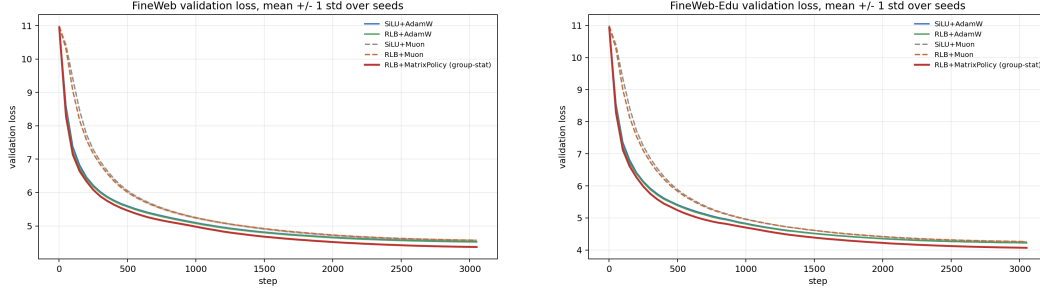


Figure 1: Mean validation-loss curves across three seeds with ± 1 standard deviation bands for FineWeb (left) and FineWeb-Edu (right).

Algorithm 1: Rational Matrix Policy step. Given gradients $\nabla \mathcal{L}(\theta^t)$ and base learning rate η_t :

1. Update exponential moving averages of rational-group pressures.
2. Compute role-depth and group-stat multipliers for each RLB matrix.
3. Update non-rational parameters with the base optimizer and update rational coefficients with the coefficient rule.
4. Update each A_l and B_l with the MatrixPolicy mixture step.
5. Apply the bounded positive-gauge rebalance to each rational group.

4 Experiments

The current evidence is a same-protocol real-corpus screen rather than the final ICLR benchmark. All rows use a 123M-parameter Transformer, GPT-2 tokenizer, 100M-token training budget, the same validation slice, three seeds (1337, 2027, 3407), equal global batch size, shared base schedule, and common evaluation cadence. The screen compares standard SiLU+AdamW, plain RLB+AdamW, RLB+Muon, and RLB+MatrixPolicy with group-stat scaling. Failed or nonfinite plain-RLB rows are counted in stability reporting.

These results are promising because the gap replicates across two web corpora and because plain RLB+AdamW is not a sufficient explanation: it is marginal on FineWeb and has seed-sensitive instability on FineWeb-Edu. They do not yet constitute a complete optimizer-paper result. A submission-ready benchmark includes tuned optimizer-family surfaces, mean-plus-standard-deviation curves, bootstrap or paired confidence intervals, divergence accounting, speed-to-target and throughput tables, academic-scale model/token scaling, transfer beyond the HPO corpora, and mechanism diagnostics that measure whether MatrixPolicy improves function-changing movement relative to gauge drift.

5 Discussion and Limitations

MatrixPolicy is a structure-specific optimizer. Its advantage, if it survives the full benchmark, should be interpreted as evidence that RLB geometry can be used productively by the optimizer, not as evidence that every rational activation or every matrix policy dominates

standard feed-forward training. The gauge analysis in Appendix B explains why generic coordinatewise adaptive updates can depend on an arbitrary RLB scale representative, but it does not prove faster convergence or lower final loss.

The main limitations are empirical. The current screen is not yet tuned against the strongest optimizer families, does not include a larger model or longer-token run, and does not yet report speed-to-target under final tuned configurations. The current confidence intervals are based on only three seeds and should be treated as stability checks. A submission-ready version needs the full optimizer benchmark described above before making final claims.

6 Conclusion

Rational feed-forward layers expose optimizer-visible structure that generic coordinatewise updates do not explicitly use. MatrixPolicy uses RLB matrix roles, group pressures, and gauge rebalancing to optimize rational FFNs with a small, localized change to the training recipe. Preliminary three-seed real-corpus results support the direction; the remaining work is to turn that evidence into a fully tuned, speed-aware, and mechanism-grounded optimizer benchmark.

References

Diederik P. Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization. In International Conference on Learning Representations, 2015. URL <https://mlanthology.org/iclr/2015/kingma2015iclr-adam/>.

A MatrixPolicy Details

Role-depth factors. For normalized layer depth $d_l = l/(L - 1)$,

$$\rho_{\text{in}}(l) = \text{clip}(1 - 0.50(d_l - 0.5), 0.55, 1.40), \quad (8)$$

$$\rho_{\text{out}}(l) = \text{clip}(1 + 1.00(d_l - 0.5), 0.55, 1.40), \quad (9)$$

$$a_{\text{role}}(l, r) = \max(0.10, 1 + 1.20(\rho_r(l) - 1)). \quad (10)$$

The local adaptive scale is

$$a_{\text{mat}}(l, r, t) = \text{clip}(3.0 a_{\text{role}}(l, r) a_{\text{stat}}(l, r, t), 0.40, 4.0). \quad (11)$$

On-policy group statistics. For group g , MatrixPolicy tracks relative gradient pressures

$$p_{\text{in},g} = \frac{\text{rms}(\nabla A_{l,g})}{\text{rms}(A_{l,g})}, \quad p_{\text{out},g} = \frac{\text{rms}(\nabla B_{l,g})}{\text{rms}(B_{l,g})}, \quad p_{\text{rat},g} = \text{rms}(\nabla \theta_{\text{coeff},g}). \quad (12)$$

With EMAs $q_{\text{in},g}$, $q_{\text{out},g}$, and $q_{\text{rat},g}$, we define

$$\pi_g = \log q_{\text{in},g} - \log q_{\text{out},g}, \quad \alpha_g = \log q_{\text{rat},g} - \frac{1}{2}(\log q_{\text{in},g} + \log q_{\text{out},g}). \quad (13)$$

Group-stat preconditioning. The group preconditioner is centered and clipped:

$$c_g = c_{\text{gain},g} c_{\text{pressure},g} c_{\text{activity},g}, \quad c_g \leftarrow c_g / \text{geomean}(c), \quad c_g \leftarrow \text{clip}(c_g, 0.75, 1.35). \quad (14)$$

The factors are

$$c_{\text{gain},g} = \left(\frac{\text{geomean}(k)}{k_g} \right)^{0.20}, \quad c_{\text{pressure},g} = \exp(0.10 d_{\text{pressure},g}), \quad (15)$$

where $d_{\text{pressure},g} = -\pi_g$ for A_l and $+\pi_g$ for B_l . The activity damping term is

$$c_{\text{activity},g} = \exp[-0.20 \text{ReLU}((\alpha_g - 0.05)/0.45)]. \quad (16)$$

Gauge rebalance. Let

$$\gamma_g = \log \text{rms}(A_{l,g}) - \log \text{rms}(B_{l,g}). \quad (17)$$

The target log-ratio is

$$\tau_g = \frac{1}{2}(\log d_g^{\text{gain}} - \log o_g^{\text{gain}}) + \beta_{\text{pressure}}(\log q_{\text{in},g} - \log q_{\text{out},g}). \quad (18)$$

The applied correction is

$$\ell_g = \frac{1}{2}(\tau_g - \gamma_g), \quad \ell_g \leftarrow \text{clip}(s(t, l) a_g^{\text{activity}} \ell_g, -\ell_{\max}, \ell_{\max}). \quad (19)$$

The transform $A_{l,g} \leftarrow e^{\ell_g} A_{l,g}$ and $B_{l,g} \leftarrow e^{-\ell_g} B_{l,g}$ preserves the homogeneous RLB function exactly and the stabilized implementation up to the RMS-floor perturbation described in Appendix B.

B Gauge Conditioning of MatrixPolicy

This appendix isolates a narrow mathematical property of the RLB parameterization and the MatrixPolicy rebalance step. The result should not be read as a dominance theorem for MatrixPolicy over AdamW. Such a theorem would be false without restrictive assumptions on the objective, initialization, hyperparameters, stochastic gradients, and implementation state. The claim established here is instead more limited: RLB blocks contain a positive scale gauge, and standard coordinatewise adaptive updates are not equivariant with respect to this gauge. The MatrixPolicy rebalance step removes part of this representative dependence by moving the matrices to a controlled point on the same gauge orbit.

B.1 RLB gauge

For one rational feed-forward layer, write

$$z_l = A_l x_l, \quad z_{l,g} = \text{group}_g(z_l), \quad (20)$$

$$r_{l,g} = \sqrt{m^{-1} \|z_{l,g}\|_2^2 + \epsilon_{\text{rms}}}, \quad u_{l,g} = z_{l,g} / r_{l,g}, \quad (21)$$

$$h_{l,g} = r_{l,g} R_{l,g}(u_{l,g}), \quad y_l = B_l \text{concat}_{g=1}^G h_{l,g}. \quad (22)$$

Here A_l selects the domains presented to the rational maps, while B_l recombines the resulting rational features. Let $B_{l,g}$ denote the block of columns of B_l corresponding to group g . For $a_g > 0$, define the groupwise transformation

$$A_{l,g} \mapsto a_g A_{l,g}, \quad B_{l,g} \mapsto a_g^{-1} B_{l,g}. \quad (23)$$

Equivalently, with $s_g = \log a_g$,

$$A_{l,g} \mapsto e^{s_g} A_{l,g}, \quad B_{l,g} \mapsto e^{-s_g} B_{l,g}. \quad (24)$$

We write this transformation as $T_s(A_l, B_l)$.

Lemma B.1 (Exact gauge in the homogeneous block). If $\epsilon_{\text{rms}} = 0$, then for any $s \in \mathbb{R}^G$,

$$f_{T_s(A_l, B_l)}(x_l) = f_{(A_l, B_l)}(x_l). \quad (25)$$

Consequently, any loss that depends on (A_l, B_l) only through the represented network function is constant on the orbit $\{T_s(A_l, B_l) : s \in \mathbb{R}^G\}$.

Proof. It suffices to consider one group and write $a = a_g$. If $A'_{l,g} = a A_{l,g}$, then $z'_{l,g} = a z_{l,g}$. When $\epsilon_{\text{rms}} = 0$, the group radius is homogeneous, so $r'_{l,g} = a r_{l,g}$. Therefore $u'_{l,g} = z'_{l,g} / r'_{l,g} = u_{l,g}$ and $h'_{l,g} = r'_{l,g} R_{l,g}(u'_{l,g}) = a h_{l,g}$. The corresponding block of B_l is scaled by a^{-1} , hence

$$B'_{l,g} h'_{l,g} = (a^{-1} B_{l,g})(a h_{l,g}) = B_{l,g} h_{l,g}. \quad (26)$$

The equality holds groupwise and therefore for the full layer output. $\square$

The implementation uses a nonzero additive stabilizer ϵ_{rms} . The exact symmetry above is then replaced by a local approximate symmetry. This distinction is minor numerically but important mathematically.

Assumption B.1 (Local nondegeneracy). On the set of activations considered, there exist constants $0 < \sigma < M < \infty$ and $S < \infty$ such that, for all groups,

$$\sigma^2 \leq m^{-1} \|z_{l,g}\|_2^2 \leq M^2, \quad |s_g| \leq S. \quad (27)$$

Assume also that the rational maps $R_{l,g}$ are bounded and Lipschitz on the corresponding compact set of normalized inputs, and that the relevant blocks of B_l have bounded operator norm.

Proposition B.1 (Approximate gauge with an RMS floor). Under Assumption B.1, there exists a constant C independent of ϵ_{rms} such that

$$\|f_{T_s(A_l, B_l)}(x_l) - f_{(A_l, B_l)}(x_l)\|_2 \leq C\epsilon_{\text{rms}}. \quad (28)$$

Equivalently, for every $\delta > 0$ there exists $\epsilon_0 > 0$ such that $\epsilon_{\text{rms}} < \epsilon_0$ implies

$$\|f_{T_s(A_l, B_l)}(x_l) - f_{(A_l, B_l)}(x_l)\|_2 < \delta. \quad (29)$$

Proof. Let $S_z = m^{-1} \|z\|_2^2$ and $a = e^s$. The radius satisfies

$$\left| \sqrt{a^2 S_z + \epsilon_{\text{rms}}} - a \sqrt{S_z + \epsilon_{\text{rms}}} \right| \quad (30)$$

$$= \frac{|1 - a^2| \epsilon_{\text{rms}}}{\sqrt{a^2 S_z + \epsilon_{\text{rms}}} + a \sqrt{S_z + \epsilon_{\text{rms}}}} \leq \frac{|1 - a^2|}{2a\sigma} \epsilon_{\text{rms}}. \quad (31)$$

The same bound, together with $S_z \in [\sigma^2, M^2]$, implies that the normalized inputs $u(z)$ and $u(az)$ differ by $O(\epsilon_{\text{rms}})$ uniformly over the compact region. Since $R_{l,g}$ is Lipschitz and the relevant blocks of B_l are bounded, the group output after applying $B_{l,g}$ differs from the original group output by $O(\epsilon_{\text{rms}})$. Summing over finitely many groups gives equation 28. The δ - ϵ statement follows by taking $\epsilon_0 = \delta/C$. $\square$

B.2 Gauge directions are not function directions

Assume first the homogeneous case $\epsilon_{\text{rms}} = 0$. Since the loss is constant along T_s , differentiating at $s_g = 0$ gives

$$\left. \frac{d}{ds_g} L(e^{s_g} A_{l,g}, e^{-s_g} B_{l,g}) \right|_{s_g=0} = 0. \quad (32)$$

Therefore

$$\langle \nabla_{A_{l,g}} L, A_{l,g} \rangle - \langle \nabla_{B_{l,g}} L, B_{l,g} \rangle = 0. \quad (33)$$

Thus $(A_{l,g}, -B_{l,g})$ is tangent to a direction that changes the parameterization but not the represented function. Equation 33 is the basic obstruction for a coordinatewise optimizer: the Euclidean coordinates contain a scale degree of freedom that is not a function-space degree of freedom.

B.3 The MatrixPolicy rebalance as a canonicalization step

MatrixPolicy uses the gauge freedom to control the relative scale of $A_{l,g}$ and $B_{l,g}$. Define the observed log-ratio

$$\gamma_g = \log \text{rms}(A_{l,g}) - \log \text{rms}(B_{l,g}). \quad (34)$$

Let τ_g be the target log-ratio computed by the optimizer from group-level statistics such as gain and relative gradient pressure. Ignoring clipping, the rebalance step is

$$\ell_g = \frac{1}{2}(\tau_g - \gamma_g), \quad A_{l,g} \leftarrow e^{\ell_g} A_{l,g}, \quad B_{l,g} \leftarrow e^{-\ell_g} B_{l,g}. \quad (35)$$

Lemma B.2 (Targeting the log-ratio). If clipping is inactive, the rebalance step in equation 35 maps γ_g exactly to τ_g .

Proof. After the rebalance,

$$\gamma'_g = \log \text{rms}(e^{\ell_g} A_{l,g}) - \log \text{rms}(e^{-\ell_g} B_{l,g}) \quad (36)$$

$$= \gamma_g + 2\ell_g = \gamma_g + \tau_g - \gamma_g = \tau_g. \quad (37)$$

□

The clipped implementation is a bounded version of the same operation. Hence the rebalance should be interpreted as a local canonicalization of the gauge representative: it changes the coordinates used by the optimizer, while preserving the represented function exactly in Lemma B.1 and up to $O(\epsilon_{\text{rms}})$ in Proposition B.1.

B.4 Coordinatewise adaptive updates are not gauge-equivariant

Let Φ denote one optimizer step applied to (A, B) and its optimizer state. A step is gauge-equivariant if, for every s ,

$$\Phi(T_s(A, B)) = T_s(\Phi(A, B)). \quad (38)$$

Gauge equivariance is a natural property here because $T_s(A, B)$ represents the same function as (A, B) in the homogeneous block. If equation 38 fails, then the optimizer's next function-space move depends on an arbitrary scale representative.

For clarity, consider a single group and suppress layer and group indices. Write the AdamW matrix update in the simplified form

$$M^+ = (1 - \eta\lambda)M - \eta \frac{m_M}{\sqrt{v_M} + \epsilon_{\text{opt}}}, \quad (39)$$

where m_M and v_M are the first- and second-moment estimates and division is coordinate-wise. Consider the log-gauge transform $A_s = e^s A$ and $B_s = e^{-s} B$. Even if we give AdamW the ideally transformed moment state,

$$m_{A,s} = e^{-s} m_A, \quad v_{A,s} = e^{-2s} v_A, \quad (40)$$

with the analogous transformation for B , the update does not commute with the gauge. Indeed, for $\epsilon_{\text{opt}} = 0$,

$$A_s^+ - e^s A^+ = \eta(e^s - 1) \frac{m_A}{\sqrt{v_A}}, \quad (41)$$

which is nonzero for $s \neq 0$ whenever $m_A \neq 0$. Similarly,

$$B_s^+ - e^{-s} B^+ = \eta(e^{-s} - 1) \frac{m_B}{\sqrt{v_B}}. \quad (42)$$

Thus a coordinatewise adaptive step is not an update on the quotient space induced by the RLB gauge. The mismatch remains generically nonzero for $\epsilon_{\text{opt}} > 0$.

B.5 A scalar separation illustrating gauge regret

The preceding non-equivariance can be made concrete in a one-dimensional example. This example is not intended to model the full MatrixPolicy implementation. It isolates the effect of using a canonical representative before a coordinatewise adaptive update.

Let

$$f(a, b) = ab, \quad L(a, b) = \frac{1}{2}(ab - y)^2, \quad (43)$$

with $a, b > 0$, $w = ab > y > 0$, and residual $\rho = w - y$. The gauge action is

$$(a, b) \mapsto (e^s a, e^{-s} b), \quad (44)$$

which preserves ab . Define the balanced representative

$$C(w) = (\sqrt{w}, \sqrt{w}). \quad (45)$$

Starting from $C(w)$, the first Adam-style normalized coordinate step with $\lambda = 0$ and $\epsilon_{\text{opt}} = 0$ has

$$a^+ = a - \eta, \quad b^+ = b - \eta, \quad (46)$$

because $\partial L/\partial a = \rho b > 0$ and $\partial L/\partial b = \rho a > 0$. The balanced representative gives

$$w_{\text{bal}}^+ = (\sqrt{w} - \eta)^2 = w - 2\eta\sqrt{w} + \eta^2, \quad (47)$$

and hence

$$L_{\text{bal}}^+ = \frac{1}{2} (\rho - 2\eta\sqrt{w} + \eta^2)^2. \quad (48)$$

In contrast, the gauge-equivalent representative

$$(a_s, b_s) = (e^s \sqrt{w}, e^{-s} \sqrt{w}) \quad (49)$$

gives

$$w_s^+ = (e^s \sqrt{w} - \eta)(e^{-s} \sqrt{w} - \eta) \quad (50)$$

$$= w - \eta\sqrt{w}(e^s + e^{-s}) + \eta^2 \quad (51)$$

$$= w - 2\eta\sqrt{w} \cosh(s) + \eta^2, \quad (52)$$

and therefore

$$L_s^+ = \frac{1}{2} (\rho - 2\eta\sqrt{w} \cosh(s) + \eta^2)^2. \quad (53)$$

Since $\cosh(s)$ is unbounded, for every $\eta > 0$ there exist gauge-equivalent representatives such that

$$L_s^+ > L_{\text{bal}}^+. \quad (54)$$

One sufficient condition is

$$\cosh(s) > \frac{\rho + \eta^2 + |\rho - 2\eta\sqrt{w} + \eta^2|}{2\eta\sqrt{w}}. \quad (55)$$

Thus the same function value can lead to a strictly worse one-step loss solely because the coordinatewise adaptive update was applied to a less balanced gauge representative.

Remark B.1 (Interpretation of the toy example). The scalar construction should be viewed as a gauge-regret example, not as a faithful model of language-model training. Its role is to show that representative dependence is not merely a formal inconvenience: it can change the one-step loss. MatrixPolicy’s rebalance step is designed to reduce this source of variation in the RLB matrices before subsequent updates accumulate optimizer state.

B.6 Robustness of the separation

The scalar separation was written in the clean limit $\lambda = \epsilon_{\text{opt}} = \epsilon_{\text{rms}} = 0$. The strict inequality persists under small perturbations.

Corollary B.1 (Local δ - ϵ robustness). Let

$$\Delta(s, \eta, \lambda, \epsilon_{\text{opt}}, \epsilon_{\text{rms}}) = L_s^+ - L_{\text{bal}}^+ \quad (56)$$

denote the one-step gap in the scalar construction, restricted to a compact nondegenerate region where optimizer denominators are bounded away from zero. Suppose that, for some s_0 ,

$$\Delta(s_0, \eta, 0, 0, 0) = m > 0. \quad (57)$$

Then, for every $\epsilon_{\text{gap}} \in (0, m)$, there exists $\delta > 0$ such that

$$|\lambda| + \epsilon_{\text{opt}} + \epsilon_{\text{rms}} < \delta \quad (58)$$

implies

$$\Delta(s_0, \eta, \lambda, \epsilon_{\text{opt}}, \epsilon_{\text{rms}}) > m - \epsilon_{\text{gap}} > 0. \quad (59)$$

Proof. On the stated compact region, the one-step maps and the loss are continuous functions of $(\lambda, \epsilon_{\text{opt}}, \epsilon_{\text{rms}})$. Since the unperturbed gap is positive, continuity gives a neighborhood of $(0, 0, 0)$ on which the change in the gap is smaller than any prescribed $\epsilon_{\text{gap}} \in (0, m)$. This proves the claim. $\square$

B.7 Scope

The analysis above supports a design motivation rather than a complete optimization proof. It establishes that RLB contains a scale gauge, that MatrixPolicy’s rebalance step acts as a bounded canonicalization of that gauge, and that AdamW-style coordinatewise adaptive steps are not invariant to the resulting equivalence relation. It does not prove that MatrixPolicy converges faster, reaches a lower loss, or dominates AdamW for arbitrary training runs. The empirical comparison in the main text must therefore remain the basis for claims about validation loss, while this appendix explains one structural reason that an RLB-specific matrix policy can behave differently from a generic optimizer.